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Class: CO-5K (B) Subject: OSY

Assignment 1.

* 2 marks:

1) Differentiate between Batch Operating system and timeshared operating system.

Batch-OS	Time-shared OS
1) Jobs are processed sequentially by the order decided by the operator in Batch OS	Jobs are processed concurrently by rapidly switching between them (time slicing)
2) It was used in first and second generation computers.	It was popular from third generation onwards.
3) It is comparatively less efficient as CPU may remain idle during I/O operations	It is comparatively more efficient as CPU ^{is shared among users} and resources are better utilized.

2) Define CPU and I/O burst cycle.

→ CPU burst cycle: It is the time period when a process is busy working with CPU.

I/O burst cycle: It is the time period when a process is busy working I/O resources.

3) State any 4 Services of Operating System.

→ Following are some of the services of Operating System:

- 1) Program execution.
- 2) I/O operations.

- 3) Protection and security.
- 4) Resource allocation.
- 5) File systems.
- 6) Accounting.

4) What is real time operating system? State its type with example.

→ A real time operating system has well defined fixed time constraints. All the processes should be executed within the defined constraints.

There are two types RTOS:

- 1) Hard RTOS: Used in medical imaging, air traffic control, etc.
- 2) Soft RTOS: Used multimedia systems, digital audio system, etc.

5) Differentiate between Multiprogrammed and Multitasking operating system.

→ Multiprogramming OS	Multitasking OS
- Concept of Context switching is used	Concept of context switching and time-sharing is used.
- Switching occurs when the execution of the current process stops	Switching occurs when the time slice of the current process is finished
- Multiprogramming increases CPU utilization by organizing jobs	Multitasking also increases CPU utilization and it also increases response time.
- Throughput is less	Throughput is moderate



6) What is System call? State its type.

→ A system call is a way for a user program to interface with the operating system. The program requests several services of the OS and the OS responds by invoking several system calls to fulfill those requests.

Types of system calls:

- 1) Process control.
- 2) File management
- 3) Device management.
- 4) Information maintenance
- 5) Communication

7) Define Process, PCB.

→ Process:

A process is a program which is under execution. It is an active entity and also known as job, task or unit of work.

PCB:

A Process Control Block (PCB) is a data structure which contains the information of the process related to it. PCB is also known as Task Control Block (TCB), entry of the process table, etc.

8) Write syntax of ps command and explain its use with the help of suitable example.

→ ps (process status) command is used to display information about the current running processes on the system.

Syntax: ps [options]

Example: ps -a → Displays process of all users.

q) Draw structure of UNIX operating system

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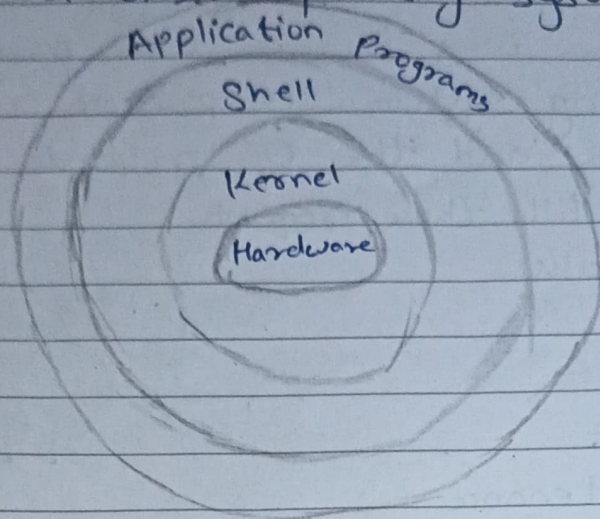


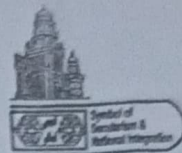
Fig: Structure of UNIX OS

* 4 marks:

1) Explain Time sharing OS.

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- Time sharing OS is a logical extension of multiprogramming which enables multiple programs to run simultaneously. Its objective is to minimize response time of process.
- In time-sharing OS, CPU executes multiple jobs concurrently by rapidly switching between them. The switches occur so frequently that each user can interact with each program it is running.
- It includes an interactive computer system which provides direct communication between user and the system.
- A time-sharing system allows multiple users to share a computer resource simultaneously by scheduling and dividing CPU's among them.
- In time-sharing OS, CPU allocates a time slice to each user. When this time is expired, the control goes on to next user.



The time allowed is extremely small and users are given the impression that each of them has their own CPU and they are the sole owners of the CPU.

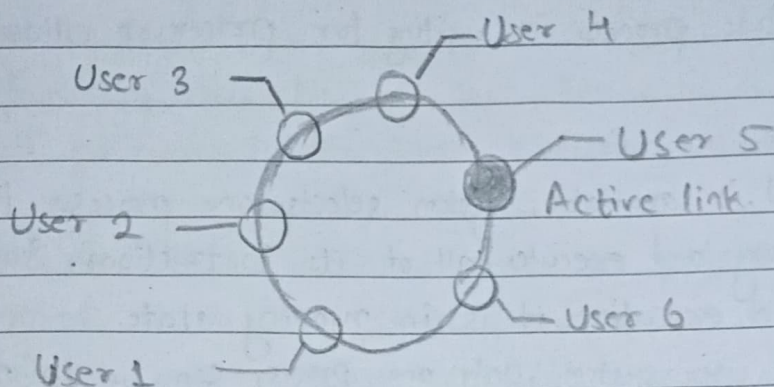


Fig: Time-sharing OS

2. With neat labelled diagram, explain process state transition.

A program is a set of instructions. A process is a program under execution and it is an active entity.

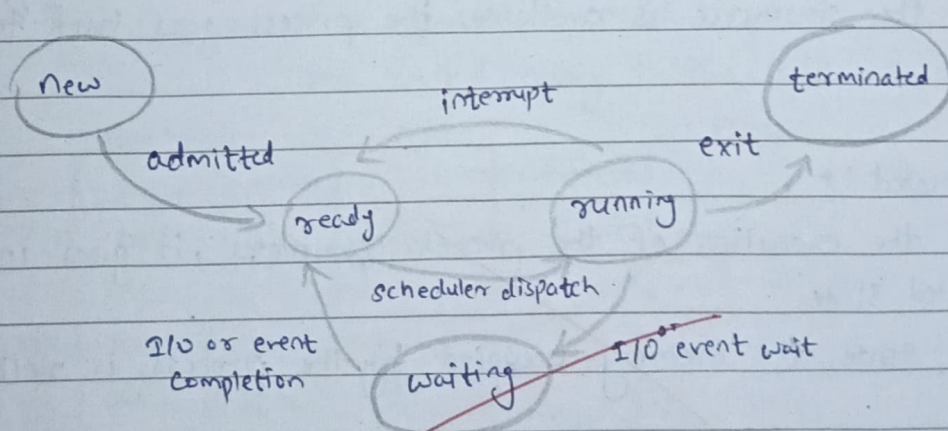


Fig: Process state transition.

Following are the states of process transition.

1) New:

When a process enters into the system, it is in new state. In this state a process is created. In this state, process is in job pool.

2) Ready:

- When the process is loaded into the main memory, it is ready for execution.
- In this state process is waiting for processor allocation.

3) Running:

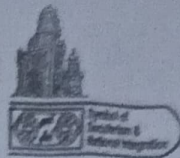
- When CPU is available, system selects one process from the main memory and executes all of its instructions. So, when a process is in execution, it is in running state.
- In single user system, only one process can be in running state whereas in multiple user system, multiple processes can be in running state.

4) Waiting:

- During execution, a process may request for an I/O resource and if it is not available, the process will go in waiting state.
- When the resource is available, the process goes back to ready state.

5) Terminated:

- When the execution of the process completes, it goes into terminated state.
- In this state, the memory occupied by the process is released.



3. With suitable diagram, describe use of following scheduling queues in processing scheduling.
- - All PCBs (Process Control Block) are kept in Process scheduling Queues by the OS.
- Each processing state has its own queue in the OS, and PCBs from all processes in the same execution state are put in the very same queue.
 - A process's PCB is unlinked from its present queue and then moved to its next state queue when its status changes.
 - The following major process scheduling queues are maintained by the Operating System:
 - Job queue: It contains all of system's processes.
 - Ready queue: The queue maintains a list of all processes in the main memory that are ready to run. This queue maintains is always filled with new processes.
 - Device queue: This queue is made up of processes that are stalled owing to the lack of an I/O device.

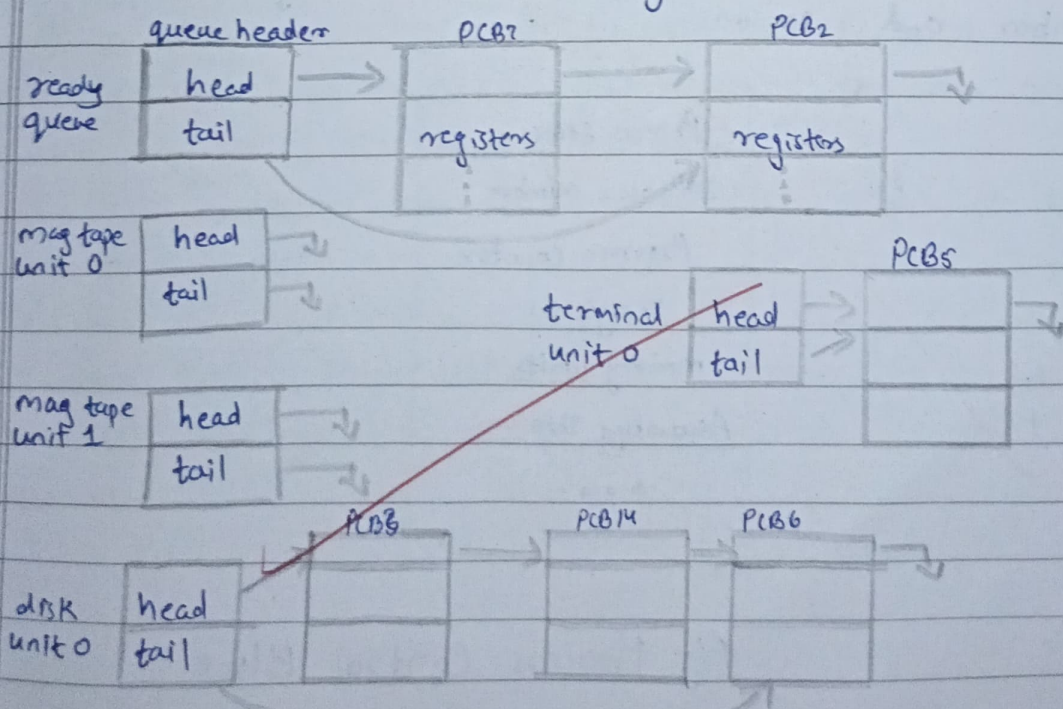


Fig: Ready queue and Device queue.

4. Describe PCB with suitable diagram.

→ Each process is represented as a process control block (PCB) in the operating system. It contains information associated with specific process.

Process state: It indicates the current state of a process.

Process number: Each process is associated with a unique number which is known as process identification number.

Program Counter: It indicates the address of the next instruction to be executed for the process.

CPU registers: The registers vary in type and number based on the computer architecture. Register includes accumulator, index registers, stack pointers and general purpose registers plus any condition code information.

Memory Management Information: It includes information such as value of base and limit registers, page tables, segment table depending on the memory system.

Accounting Information: This information includes the amount of CPU used, time limits, account holders, job or process number and so on.

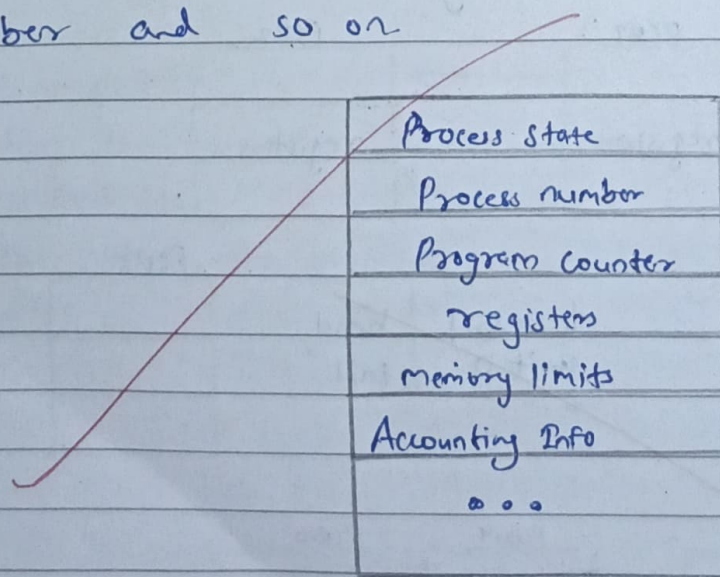
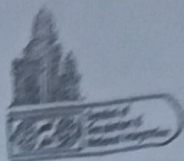


Fig: Process Control Block.



5. Define the term inter process communication (IPC). Explain any one technique of IPC.

→ Interprocess Communication is the mechanism provided by the Operating System that allows processes to communicate with each other. Cooperating processes require an Inter-process communication mechanism that will allow them to exchange data and information.

Shared Memory:

- When a process creates a shared memory segment, it makes a region of its address space accessible to other processes. Any other process that needs to communicate with the creator can access this shared region.
- For any process to use a shared memory segment, it must connect to that segment and map it into its own address space. This allows the process to access the shared data.
- This allows multiple processes to read from write to the same memory location, facilitating their inter-process communication.
- The form of data and the location are determined by these processes who wants to communicate with each other.
- They are also responsible for ensuring they are not writing to the same location simultaneously.
- These processes are not under the control of the operating system.
- Once a shared memory segment is set up, the operating system's kernel steps back and all subsequent reads and writes to the shared memory are treated as normal memory access.

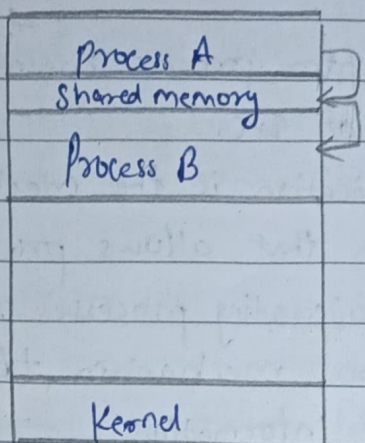


Fig: Shared Memory IPC

- 6- Explain working of CPU switch from process to process with neat labelled diagram.
- - A CPU switching from one process to another process is referred to as context switch.
- Context switch is a mechanism that store and restore the state or context of a CPU in Process Control Block so that a process execution can be resumed from the same point at a later time.
 - When the scheduler switches the CPU from one process to another process, the context switch saves the contents of all process registers for the process being removed from the CPU, in its process control block.
 - Context switch includes two operations; state save and state restore. State save operation stores the current info of running process into PCB. State restore operation restores the information of process to be executed from its PCB.
 - Switching the CPU from one process to another process involves performing the state save operation for the currently executing process and a state restore operation for the process to be executed.

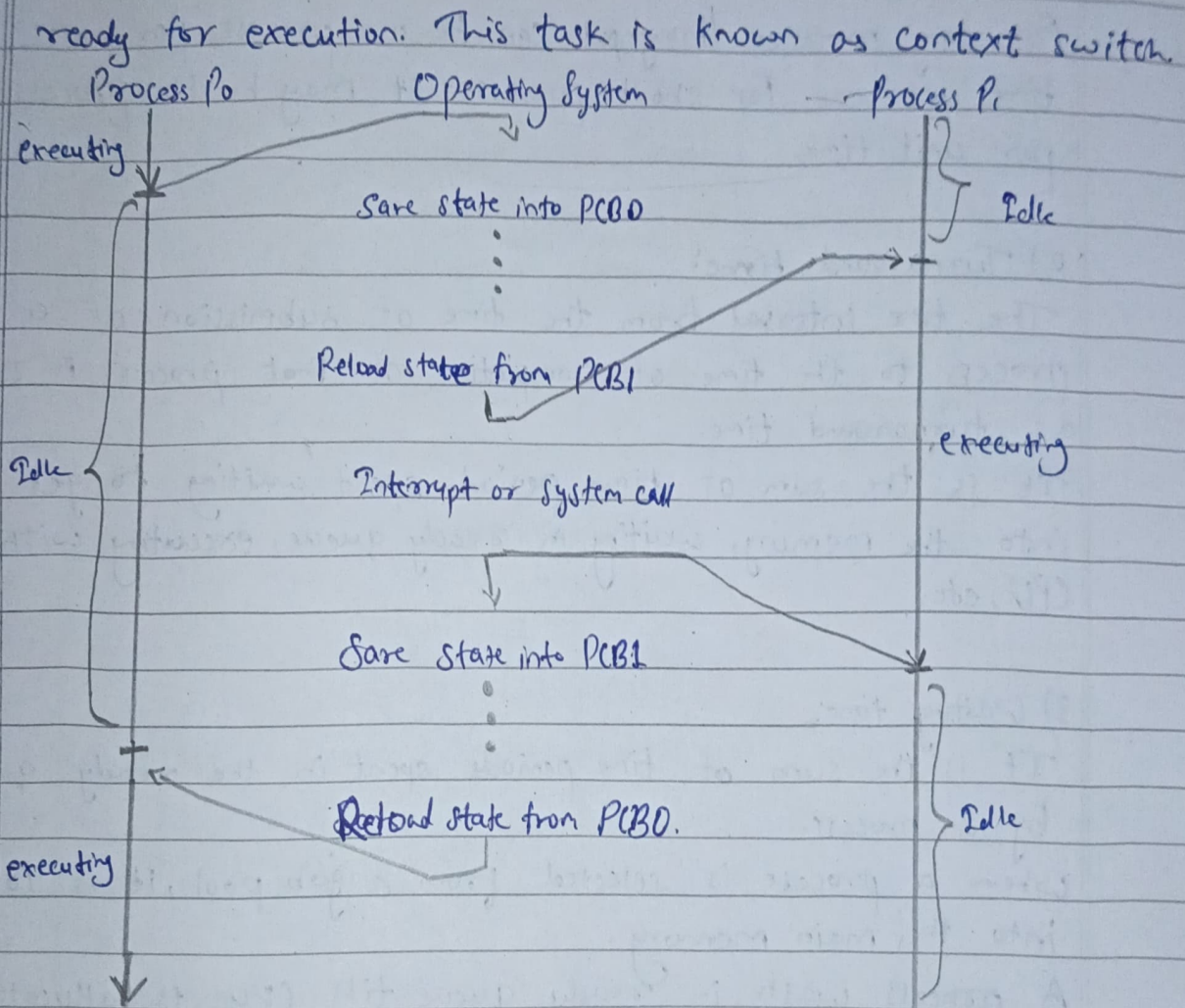
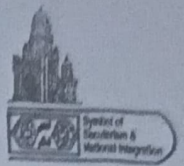


Fig: Context Switch

Describe different scheduling criteria.

1) Throughput:

It is the number of processes that are completed per unit time.

It is a measure of work done in the system.

When CPU is busy in executing processes, then its work is being done in the system.

Throughput depends upon the execution time required by the processes.

For long processes throughput can be one process per unit time whereas for short processes it may be 10 processes per unit time.

2) Turnaround time:

The time interval from the time of submission of a process to the time of completion of that process is called as turnaround time.

It is the sum of time period spent waiting to get into the memory, waiting in ready queue, executing with CPU, etc.

3) Waiting time:

It is the sum of time periods spent in the ready queue by a process.

When a process is selected from a job pool, it is loaded into the main memory.

A process waits in ready queue till CPU is allocated to it.

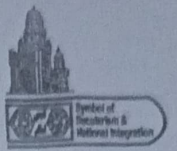
Once the CPU is allocated to the process, it starts its execution and it requests for resources.

When the resources are not available that process goes back to ready queue.

In ready queue again it waits for CPU allocation.

4) Response time:

The time period from submission of a request until the first response is produced is called as response time.



It is the time when system responds to the process request not the completion of a process.

In the system, a process can produce some output fairly early and can continue computing new results while previous results are being output to user.

8. Difference between user level and Kernel level thread

	User level thread	Kernel level thread
1)	User-level threads are faster to create and manage	Kernel level threads are slower to create and manage.
2)	Implementation is by a thread library at the user level	Implementation is done by the Operating System
3)	User-level thread is generic and can run on any OS	Kernel level thread is specific to the OS.
4)	Multi-threaded applications cannot take advantage of multiprocessing	Kernel routines themselves can be multithreaded.

10. Explain Multiprogramming OS.

In multiprogramming, more than one program lies in the memory. The scheduler selects jobs to be placed in ready queue from a number of programs.

The ready queue is placed in memory and the existence of more than one program in main memory is known as multiprogramming. Since there is only one processor, these multiple programs cannot be executed at a time.

Instead the operating system executes part of one program, then the part of another and so on.

Example: Users can open word, excel and other applications in System

- Single program:

Run Wait Run Wait

Time →

- Multiprogramming with two programs:

Program A → Run Wait Run Wait

Program B → Wait Run Wait Run

Combined → RunA RunB Wait RunA RunB

Fig: Multiprogramming OS

9. Differentiate between multitasking and multiprogramming.

→ Multiprogramming OS	Multitasking OS.
i) Concept of Context switching is used	Concept of context switching time sharing is used.
ii) Switching occurs when the process currently running stops	Switching occurs after the time slice of current process completes.
iii) It uses job scheduling algorithms so that more than one program can run at same time.	Time sharing mechanism is used so that multiple tasks can run at the same time.
iv) Execution of process takes more time	Execution of process takes less time.
v) Throughput is less	Throughput is moderate
vi) Efficiency is less	Moderate efficiency



* 6 marks:

1. Compare between Command Line and Graphical user interface.	
Command Line Interface (CLI)	Graphical User Interface (GUI)
i) Interaction is done by typing commands.	Interaction with devices is by graphics and visual components and icons.
ii) Commands need to be memorized	Visual indicators and icons are easy to understand
iii) Less memory is required for storage.	More memory is required for storage.
iv) Use of keyboard makes CLI quicker	Use of mouse for interaction makes it slow.
v) Highly accurate	Less accurate compared to CLI.
vi) CLI does not change, remains same over time	Structure and design of GUI can change with the updates.

List the different types of OS. Explain Multiprocessor OS with advantages and disadvantages.

Following are the different types of OS:

- 1) Batch OS.
- 2) Multiprogramming OS.
- 3) Multiprocessor OS.
- 4) Multitasking OS.
- 5) Realtime OS.
- 6) Mobile OS.

Multiprocessor OS:

- Multiprocessor systems are also known as parallel systems tightly coupled systems.
- These systems have two or more processors in close communication and they share computer resources such as bus, peripheral devices, clock, memory, etc.
- The whole task of multiprocessing is managed by the operating system, which allocates different tasks to be performed by the various processors in the system.
- Applications designed for the use in multiprocessing are said to be threaded, which means that they are broken into smaller routines that can be run independently.
- Multiple CPUs are linked together so that a job can be divided and executed more quickly.
- When a job is completed, the results from all CPUs are combined to provide final output.
- There are two types of multiprocessor OS: 1) Symmetric multiprocessor OS.
2) Asymmetric Multiprocessor OS.

Advantages:

- 1) Increased reliability: If one processor fails, others can take over.
- 2) Higher throughput: Multiple processors can complete more tasks simultaneously.
- 3) Improved Scalability: You can add more processes to handle the workload.

Disadvantages:

- 1) Increased complexity: The OS is more complex.
- 2) Higher Cost: Initial cost is higher.
- 3) Complex Software: Writing software to use multiprocessor OS is challenging.

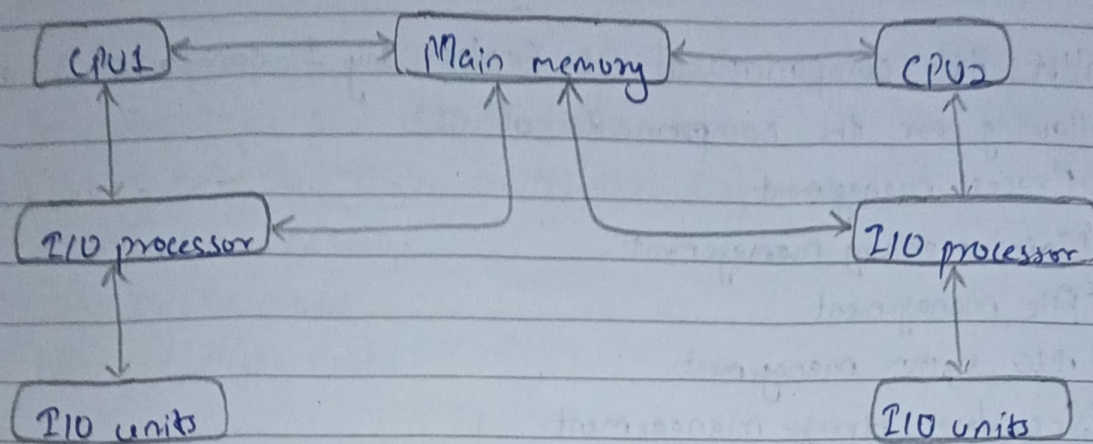


Fig: Multiprocessor OS.

Distinguish between Long term scheduler and short term scheduler.

Long term scheduler	Short term scheduler
i) It is job scheduler	It is CPU scheduler.
ii) It selects processes from job pool and loads them into main memory.	It selects processes from ready queue which are ready to execute and allocates CPU to them.
iii) Access job pool and ready queue	Access ready queue and CPU.
iv) It executes much less frequently.	It executes more frequently.
v) It is slower than the Short term scheduler	It is faster comparatively.
vi) It controls the degree of multiprogramming.	It provides lesser control over degree of multiprogramming.

4. Enlist the components of OS. Explain any 1 component in detail.
→ Following are the components of OS:

- 1) Process management
- 2) Main memory management
- 3) File management
- 4) I/O system management
- 5) Secondary storage management

- Main memory management:

- Main memory is a large array of words or bytes, ranging in size from hundreds of thousands to billions.

- Each word or byte has its own address. Main memory is a repository of quickly accessible data shared by CPU and I/O devices.

- The central processor reads instructions from main memory during the instruction fetch cycle and both reads and writes data from main memory during the data fetch cycle.

- The main memory is generally the only large storage device that the CPU is able to access and address directly.

- The operating system is responsible for the following activities in connection with main memory management:

- Keeping track of which parts of memory are currently being used and by whom.

- Deciding which processes and data to move into and out of memory.

- Allocating and deallocating memory space as needed



5. List the different multithreading models. Explain multi-threading with advantages and disadvantages.

→ Following are the three multi-threading models.

1) Many-to-One model.

2) One-to-One model

3) Many-to-Many model.

1) Many-to-One model:

The many-to-one model maps many user-level threads to one kernel level thread.

Thread management is done by the thread library in user space; so it is efficient but the entire process will block if a thread makes a blocking system call.

Also, because only one thread can access the kernel at a time, multiple threads are unable to run in parallel on multiprocessors.

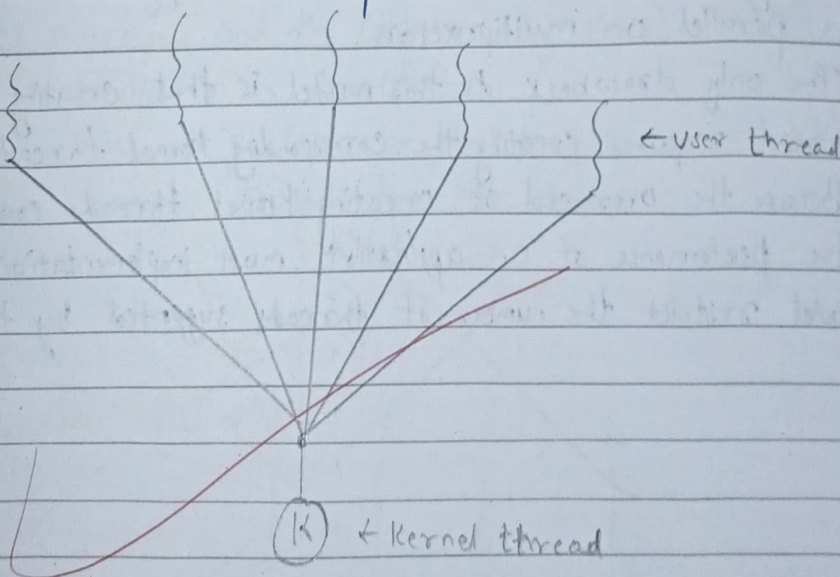


Fig: Many-to-One model.

Advantages:

- Mainly used in language system, portable libraries.
- One kernel thread controls multiple user threads.

Disadvantages:

- Parallelism is not supported by this model.
- One block can block all user threads.

2) One-to-One Model:

- The one-to-one model maps each user thread to a kernel thread.
- It provides more concurrency than the many-to-one model by allowing another thread to run when a thread makes a blocking system call; it also allows multiple threads to run in parallel on multiprocessors.
- The only drawback to this model is that creating a user thread requires creating the corresponding kernel thread.
- Because the overhead of creating kernel threads can burden the performance of an application, most implementations of this model restrict the number of threads supported by the system.

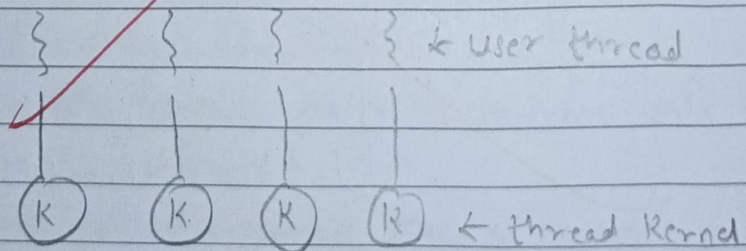


Fig. One-to-One model.



Advantages:

- + More concurrency because of multiple threads can run in parallel on multiple CPUs
- Less complication in the processing.

Disadvantages:

- Kernel thread is an overhead
- Limited number of threads.

3) Many-to-many model:

- The many-to-many model multiplexes many user level threads to a smaller or equal number of threads (kernel).
- The number of kernel threads may be specific to either a specific application or particular machine.
- In many-to-many model, developers can create as many user threads as necessary and the corresponding kernel threads can run in parallel in a multiprocessor.
- Also when a thread performs a blocking system call, the kernel can schedule another thread for execution.

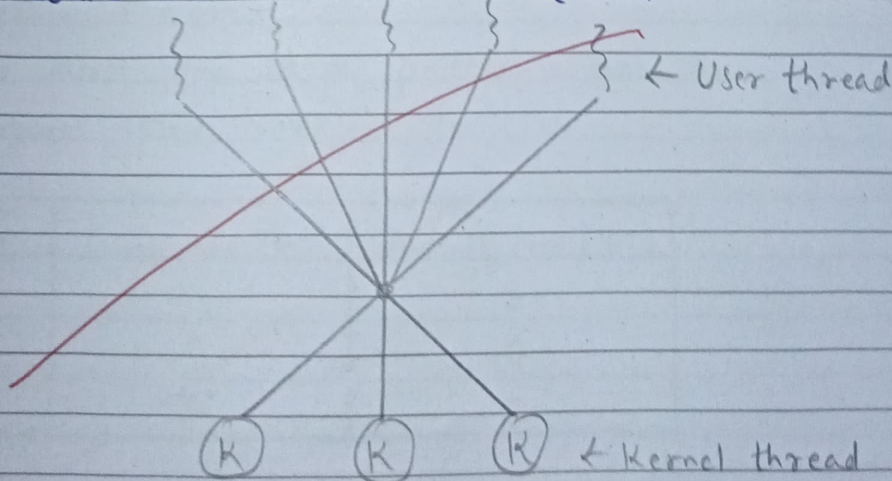


Fig: Many-to-many model

Advantages:

- Many threads can be created as per user's requirement
- Multiple kernel or equal to user threads can be created

Disadvantages:

- True concurrency cannot be achieved
- Multiple threads of kernel is an overhead for operation

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